information to improve trading strategies and to lower execution costs.[1980] To the extent that the advantage of having this information depends on other traders not having it, this economic effect will represent a transfer from the current users of depth of book information to those market participants who will now get access to, and will be able to utilize, this information.[1981] In particular, a more widespread dissemination of depth of book information may cause market prices to adjust to this information more rapidly as more people react to this information. Once market prices settle to a level that reflects this information, the opportunity to profit from having additional depth of book information may be lost.

The Commission believes that market participants who use strategies related to order routing, order placement, and order execution, may benefit from the new depth of book information, provided that currently they do not already obtain this information via proprietary data feeds. For instance, market participants may seek to get priority in the queue at a particular price level behind the top of book by posting a limit order. Such a strategy may benefit from being able to see the depth at these price levels at multiple exchanges in order to evaluate which exchange's queue would provide the order with the highest execution priority. To the extent this is the case, the Commission believes that market participants who previously did not have access to additional depth of book information will benefit by being able to better run such strategies. This could improve order execution quality for these market participants (or their clients).[1982]

1980 Commenters agreed including depth of book information in core data would help lower execution costs. See, e.g., RBC Letter at 4; ICI Letter at 8–9.

1981 See infra Section V.C.1(c)(iv).

1982 For a discussion of order execution quality and the provision of execution services by broker-dealers, see supra Section V.B.3(e).

The Commission believes that many of the market participants who utilize such strategies already have access to full depth of book information via subscriptions to proprietary feeds; for these traders the additional core data will not produce a direct benefit.[1983] The Commission is unable to quantify the number of market participants who currently run these types of strategies without using depth of book information because the Commission does not have access to information on specific strategies utilized by individual traders in the market.[1984]

Also, the Commission believes that there may be market participants that would start running these order routing strategies if the data were available to them at prices lower than the current prices for equivalent data in proprietary feeds.[1985] These market participants might currently find that the value of attempting such strategies without DOB data is too low to justify them, but that access to additional DOB data through these elements of the new definition of

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1983 The inclusion of depth of book information may benefit those market participants who already use depth of book information by providing alternatives to proprietary feeds. For a discussion of this effect, see infra Section V.C.1(c)(iv). Also, the Commission understands that some market participants who use proprietary feeds as their main source of market data also use the exclusive SIP feeds as a backup. For such market participants, the expansion of DOB information may improve the value of a core data feed as a backup.

1984 The Commission requested comment on market participants who run order routing strategies without access to DOB information but did not receive information from commenters that would help quantify the number of market participants that use such strategies. The Commission believes that it is possible that the inclusion of this information in the definition of core data, along with reductions in the latency differential that will result from the decentralized consolidation model, may benefit market participants who do not currently run these strategies but who will choose to start running them as a result of the changes. For more discussion on this possibility, see infra Section V.C.4(b).

1985 See infra Section V.C.2(b)(i) for a discussion of consolidated market data fees and Section V.C.4(b) for a discussion of market participants who may start running such strategies.

core data will enable them to run such strategies effectively. To the extent that such market participants exist, the additional DOB data will be a benefit to them as well.

The revision in trading strategies discussed above may result in changes to the decisions traders make about where to route their orders among the various trading venues. Market participants may find that depth of book information suggests trading opportunities on exchanges to which they would not have otherwise routed their orders. The Commission is uncertain about the magnitude of this effect or which trading venues may gain or lose order flow as a result. The Commission cannot determine how many market participants may choose to change routing strategies as a result of the new depth of book information, nor to what extent the new depth of book information will cause market participants to change where they route their orders.[1986]

Also, the Commission believes that the more widespread dissemination of depth of book information may result in more efficient pricing.[1987] As more traders take advantage of information contained in the depth of book data, prices will reflect this information more quickly.[1988] Therefore, more widespread dissemination of depth of book information may lead to pricing that better reflects available information. The size of this effect depends on the willingness and ability of market participants who currently rely solely on SIP data to make use of the information in the new depth of book data, which is unobservable.

1986 One commenter stated that this information "should be essential" to the Commission's analysis, yet did not provide such information. See Nasdaq Letter IV at 47. The Commission requested comment on this issue but did not receive information to help determine these effects, which is unobservable in the current market.

1987 For further discussion of this point, see infra Section V.D.1.

1988 A commenter agreed that including depth of book information in core data would improve price discovery. See RBC Letter at 4.

The Commission believes that there may be gains from trade that will be realized as a result of adding this depth of book information as an element of core data. The possibility for this benefit to materialize relies on the extent to which there exist market participants who will be willing to send orders that "walk the book"[1989] but currently do not do so because they do not see what is beyond the top of the book. This situation represents a current economic inefficiency because there are potential gains from trade that are not realized because of a lack of information. This would benefit both the market participant walking the book and the market participants who posted orders behind the BBO that will be filled as a result of the trade.

Relatively few orders actually execute at prices outside the NBBO,[1990] which implies that trading against quotes away from the NBBO on a single exchange, using a single marketable order, does not occur frequently. In addition, an analysis of a sample of trading in ten stocks on the Nasdaq exchange found that an average of 0.65% of market orders walked through the best displayed price level for these ten stocks.[1991] Therefore, the Commission believes that there may be limited benefits from additional DOB information in the particular hypothetical case of market participants who currently rely solely on SIP data for market information and who will submit market orders to trade against limit orders beyond the top of the book on a single exchange when the depth of book information is available in core data. However, the size of the

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1989 See supra note 1673.

1990 See id.

1991 See Nikolaus Hautsch and Ruihong Huang, *Limit Order Flow, Market Impact and Optimal Order Sizes: Evidence from NASDAQ TotalView-ITCH Data*, at 10, Table 3 (Aug. 22, 2011), available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1914293 (Retrieved from SSRN Elsevier database).

benefit depends on the willingness of market participants to walk the book after receiving the new DOB information, as well as their trading interest, and this is unobservable in the current market.[1992]

The addition of depth of book information to expanded core data will increase the total message traffic in expanded core data, and this increase in message traffic will be accompanied by costs to market participants to set up the infrastructure required to handle this new level of traffic. Additionally, competing consolidators and SROs may incur implementation costs related to receiving and generating the information necessary to process and disseminate consolidated market data. However, market participants are not required to receive (or display) the DOB information contained in expanded core data, and competing consolidators will not be required to disseminate all of the information in consolidated market data, including DOB information contained in expanded core data, so they will not incur these costs unless they choose to receive or disseminate this information.[1993] These costs are discussed below in Section V.C.1(c)(iv).

(iii)Effects of Addition of Auction Information

The Commission is adding "auction information" as an element of core data. This will result in all auction information currently disseminated by exchanges via proprietary data feeds being made available to subscribers of these elements of core data feeds. This will have effects that include changes to market participants' trading strategies, gains from trade as a result of new

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1992 The Commission requested comment on to what extent any benefits of including depth of book information in core data depend on the degree to which orders walk the book. No commenters provided information on the willingness of market participants to walk the book if they received the new DOB information from expanded core data.

1993 See supra Sections II.C.2(a) and III.C.8(a)(ii).

participation in auctions, potential improvements to price discovery in auctions, changes to order routing decisions, and a significant reduction in the value of dedicated proprietary auction feeds.

Several commenters stated that auction information may only be useful to sophisticated investors who already receive it and that including it in core data may not benefit most market participants.[1994] However, other commenters stated that investors, including retail investors, would benefit from including auction information in expanded core data.[1995] The Commission disagrees with the first set of commenters. The Commission believes including auction information in core data would expand its availability and allow more market participants to receive the benefits described below. Even if market participants do not directly access auction information, including it in core data may indirectly benefit market participants. If broker-dealers that do not currently receive auction information utilize the auction information included in core data to improve their handling of customer orders that participate in opening and closing auctions, it may improve their execution quality. More participation in closing auctions could also improve the price efficiency of closing prices, which could lead to better trading outcomes for market participants that rely on closing prices resulting from closing auctions, but do not participate directly in closing auctions.

As discussed above, some auction information is currently available to market participants through specialized feeds,[1996] and also a limited set of auction information is

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1994 See, e.g., TD Ameritrade at 5; Nasdaq Letter IV at 33; Data Boiler Letter at 31.

1995 See, e.g., CBOE Letter at 21; Angel Letter at 8; SIFMA Letter at 7; Virtu Letter at 5.

1996 See supra Section V.B.2(a).

available through the current SIP feeds.[1997] The availability of these feeds enables access to a limited set of auction information for some market participants without having to subscribe to full DOB feeds. To the extent that any market participants find these specialized auction feeds sufficient for their trading needs, the Commission believes that the addition of all auction information as an element of core data will have a limited effect on these market participants.[1998] To the extent that those market participants make up a large share of the market participants who would be interested in using additional auction information, the Commission believes that the effect of adding auction information may be limited.[1999] The Commission believes that the extent of this limitation is reduced by the fact that not all auction information is available to market participants through such feeds. The Commission does not have data on the number of market participants with these proprietary feed subscriptions.

The Commission believes that auction information contains insights useful to market participants in devising and executing trading strategies.[2000] Therefore, the Commission believes that adding this information as an element of core data will benefit those market participants (including retail investors) who currently do not access such information, as well as their clients.

1997 See id.

1998 Market participants who currently receive auction information through proprietary feeds may switch to using the auction information contained in expanded core if it is available at lower cost than equivalent data from proprietary feeds. See infra Section V.C.2(b).

1999 Since the cost to integrate multiple auction feeds into a single feed is a fixed cost in producing a market data feed, the Commission believes that there would still be a benefit from the rule in the form of competing consolidator integrated auction feeds, which could be cheaper for market participants than integrating the feeds themselves.

2000 See Proposing Release, 85 FR at nn.344–46. Commenters agreed auction information is useful for predicting price movements and placing orders in closing auctions. See supra note 1678.

To the extent that these market participants can use this auction information, the addition of this information as an element of core data will enable them to produce better trading strategies and lower execution costs for their own orders and for their clients' orders, as well as facilitate best execution.[2001] To the extent that the advantages of possessing auction information come from exploiting the trading decisions of market participants who lack this information, this effect will represent a transfer from those market participants who currently have auction information to those market participants who would obtain access to it through this rule and are able to exploit it to improve their trading strategies.[2002] The Commission believes that this auction information could potentially be used across all trading venues, including exchange auctions, continuous exchange trading, and off-exchange venues.

The Commission believes that the addition of auction information as an element of core data will result in increased participation in auctions, which may allow market participants to realize potential gains from trade. Commenters suggested that there are market participants who do not currently trade in auctions because they do not access auction data due to the cost of

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2001 Commenters agreed that including auction information in core data may promote more informed and effective trading in auctions. See, e.g., Clearpool Letter at 15.

2002 One commenter agreed that including auction information in core data would level the playing field for investors. See Virtu Letter at 5. Commenters also agreed that including auction information in core data would reduce information asymmetry between subscribers of SIP data and proprietary DOB feeds. See, e.g., IntelligentCross Letter at 4; Clearpool Letter at 15. One commenter stated including auction information in core data would benefit retail investors by reducing information asymmetry between retail investors and more informed market participants. See Angel Letter at 8 ("Retail investors should be properly informed with appropriate information about the indicative auction price and the trading imbalance. Otherwise, we will be at a serious disadvantage to other better informed players.").

proprietary feeds.[2003] To the extent that such market participants exist, including auction information in core data will allow these market participants to access this information, which may allow them to gain insights about trading opportunities that induce them to trade in auctions.[2004] Commenters stated that an increase in auction participation will also increase auction liquidity.[2005] The Commission agrees and believes that increased auction liquidity will also result in increased trading during auctions, which could benefit both sides of the trade, thus resulting in an economic benefit. To the extent that market participants who start trading in auctions as a result of gaining access to auction information possess insights beyond what can be inferred from auction information, increasing the number of participants in auctions as described above should improve price discovery in the auction process.[2006] The Commission believes that those who do not participate in auctions because they do not access auction information are unlikely to possess insights beyond what can be inferred from auction information. This is because any market participant who has such insights would find it worthwhile to purchase auction information and participate in the auction so as to exploit the value of the insights. Therefore, the size of this effect depends on the number of market participants who currently possess such insights relative to those who do not who start participating in auctions as a result of this rule and the size of their resultant auction trades. Both of these effects are unobservable in the current market.

2003 Commenters stated that the costs of proprietary data feeds prevent some market participants from competing in auctions. See, e.g., ICI Letter at 9–10; SIFMA Letter at 7.

2004 Commenters agreed that including auction information in core data would result in more market participants participating in auctions. See, e.g., BlackRock Letter at 2; ICI Letter at 9–10.

2005 See, e.g., BlackRock Letter at 2; IntelligentCross Letter at 4.

2006 Commenters agreed that adding auction information to core data would improve price discovery. See, e.g., BlackRock Letter at 2; ICI Letter at 9–10; Data Boiler Letter at 31.

Further, the Commission believes that the addition of auction information as an element of core data may affect the order routing decisions of market participants who currently do not have access to auction information. For example, some off-exchange trading venues cross market-on-close orders before the closing auction takes place and later settle the trades at the closing auction price. To the extent auction information is made available prior to the applicable cut-off time, if any, for the submission of closing orders to off-exchange venues, having access to auction imbalance information may affect market participants' decision to route a closing order to either an off-exchange venue or to the closing auction on the primary listing exchange. For example, a market participant who gets access to auction information through a subscription to these elements of new core data might decide not to route the order to an off-exchange venue so as to be able to participate in the auction using the new information available. Additionally, this auction information could also affect decisions made during the time when auction information is disseminated about whether an order should participate in continuous market trading or an auction. For example, if auction imbalance information indicates that an order would have a low probability of executing in an auction (or would be likely to execute at a worse price than if the order executed during continuous trading), then a market participant may decide the order should participate in continuous market trading, instead of the auction, to increase the chance the order is filled (or executed at a better price).[2007] However, the overall effect of

[2007] Similarly, if auction imbalance information indicated a market participant's order would be more likely to execute in an auction at a better price, then the market participant may choose to have the order participate in the auction instead of continuous trading.